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E- and chemoselective thia-Michael addition to benzyl allenoate

Rifhat Bibi[a], Amna Murtaza[a], Khalid Mohammed Khan[b], Zia ur Rehman[a], Aamer Saeed[a], Muhammad Nawaz Tahir[c], and Abbas Hassan[a]

[a]Department of Chemistry, Quaid-i-Azam University, Islamabad, Pakistan; [b]HEJ Research Institute of Chemistry, International Center for Chemical and Biological Sciences, University of Karachi, Karachi, Pakistan; [c]Department of Physics, Sargodha University, Sargodha, Pakistan

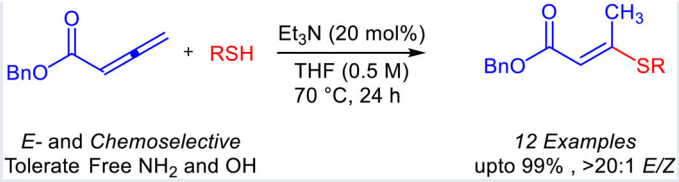
ABSTRACT

Different thiols were successfully reacted with benzyl allenoate resulting in *E*-selective thia-Michael addition product with α,β-unsaturation as confirmed by single crystal x-ray crystallographic analysis. The thia-Michael addition is chemoselective and free amine and alcohol groups were well tolerated. Catalytic triethylamine was required for high conversion. Fair to excellent yields were obtained for a variety of aliphatic, aryl and heteroaryl thiols.

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KEYWORDS
Thia-Michael addition; α-β-unsaturation; E-selective; β-substituted thioenoates; chemoselective

GRAPHICAL ABSTRACT



1. Introduction

Michael addition reaction is a widely utilized and historically important reaction in organic synthesis.[1] It is more than a century old reaction in which addition of soft nucleophiles to electron deficient alkenes occurs.[2] The nucleophiles consist of stabilized enolates, organocuprates, and oxygen, nitrogen and sulfur anion or pronucleophiles. The electrophiles include unsaturation attached to electron withdrawing groups such as enone, enoate, enal, nitroalkene, alkene nitrile and analogous alkyne compounds. Sulfur containing compounds are considered to be soft nucleophiles and the thia-Michael addition reactions are well documented.[3] α,β-Unsaturated carbonyl compounds demonstrated their higher ability as Michael acceptor when reacted with thiols.[4] Similar reactivity have been utilized to a great extent for the annulation and the synthesis of diverse heterocyclic structures.[5]

Mostly the reaction reported for thia-Michael addition to allenoate and alkynoates result the *Z*-isomer as main product.[6] For instance, the thia-Michael addition to ethyl 2-methylbuta-2,3-dienoate have been reported to give mixture of products in the absence of any additive, favoring the conjugated *Z*-selective Michael addition product.[7] This is, however, not the case when PBu_3 was utilized as the catalyst and only the non-conjugated Michael addition product is observed (Scheme 1, reaction 1). Recently, a nonselective thia-Michael addition of thiophenol to methyl 2-butynoate in the presence of sodium hydroxide has been reported to give 1:1 mixture of *E*/*Z* isomers.[6c] Interestingly, the resulted β-substituted thioenoates upon further reaction with *in situ* generated alkyl cuprates, afforded the thiol-substituted products in stereospecific fashion (Scheme 1, reaction 2). This methodology has been applied in the total synthesis of juvenile hormones.[6a]

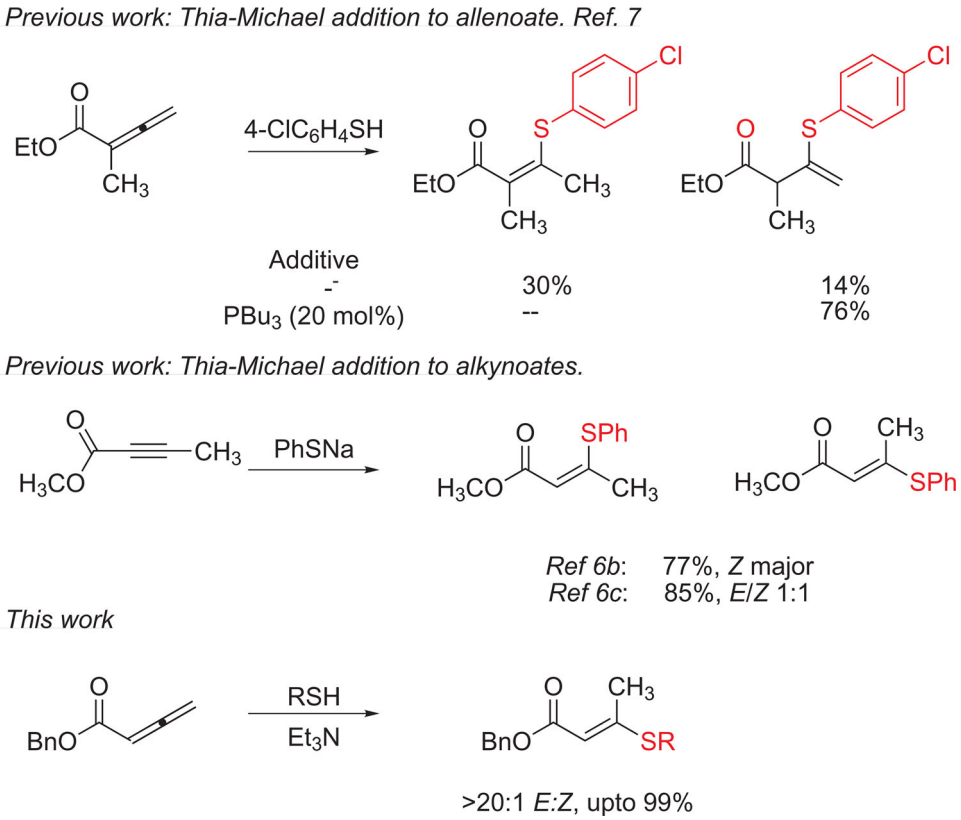
Owing to the importance of the olefin selectivity in thia-Michael addition, herein, we report a general and mild *E*-selective and chemo-selective thia-Michael addition to benzyl allenoate as surrogate for alkynoate.

2. Results and discussion

In a typical reaction, when thiophenol **1a** was treated with benzyl allenoate in THF with 150 mol% of triethylamine, it resulted in benzyl β-phenylthio-2-butenoate **2a** in 60% yield as single *E*-isomer selectively. Lowering the amount of triethylamine increased the yield and in the presence of only 20 mol% of triethylamine, 72% of **2a** was isolated (Table 1, entries 1 and 2).

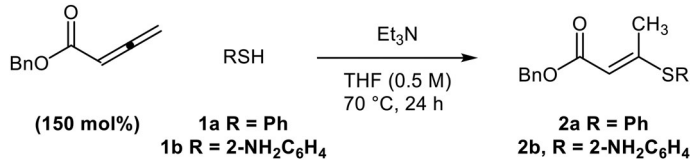
The catalytic amount of triethylamine was found to be optimal for other examples as well. When 2-aminothiophenol **1b** was treated with benzyl allenoate and 20 mol% of triethylamine, (*E*)-benzyl 3-(2-aminophenylthio)but-2-enoate **2b** was isolated in 68% yield. As the substrate contains an *ortho* amino group, it was assumed that the reaction will not require any additional triethylamine. Surprisingly, with only

CONTACT Abbas Hassan ahassan@qau.edu.pk Department of Chemistry, Quaid-i-Azam University, Islamabad 45320, Pakistan.
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Scheme 1. Thia-Michael addition reaction and its utility in stereospecific transformation.

Table 1. Triethylamine optimization for thia-Michael addition to benzyl allenoate.



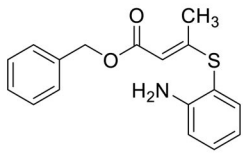
(150 mol%) **1a R = Ph** **1b R = 2-$NH_2C_6H_4$** **2a R = Ph** **2b, R = 2-$NH_2C_6H_4$**

S. No.	RSH	Et_3N (mol %)	Yield (%) (*E*:*Z*)
1	PhSH	150	60% (>20:1)
2	PhSH	20	72% (>20:1)
3	2-$NH_2C_6H_4SH$	20	68% (>20:1)
4	2-$NH_2C_6H_4SH$	10	60% (>20:1)
5	2-$NH_2C_6H_4SH$	–	26% (>20:1)
6	2-$NH_2C_6H_4SH$	150	92% (>20:1)

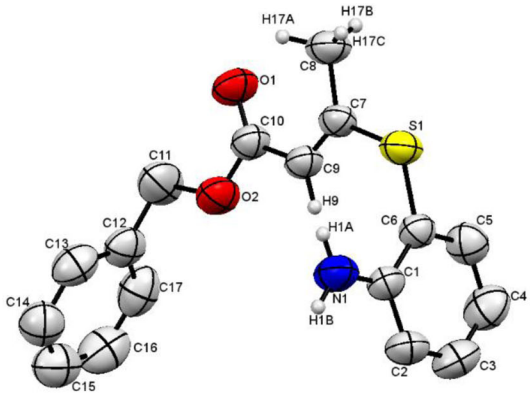
10 mol% of triethylamine, **2b** was isolated in 60% yield while, in the absence of triethylamine only 26% of **2b** was isolated (Table 1, entries 4 and 5). This suggested that a catalytic amount of triethylamine acts as proton shuttle.[8] However, in the presence of 150 mol% of triethylamine, **2b** was isolated in much higher 92% yield as single *E*-isomer.

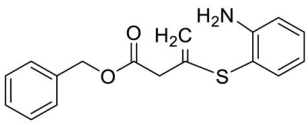
In the case of **2b**, it was important to confirm the chemoselectivity for thia *vs* aza-Michael addition and the configuration of the double bond. Although NMR analysis was useful, single crystal X-ray analysis confirmed the structure of **2b**. The product was indeed thia-Michael adduct with *E*-enone geometry (Figure 1). We were unable to detect any *Z*-**2b** from the NMR analysis of the crude reaction mixture. Interestingly, when the reaction was carried out in the absence of triethylamine, (Table 1, entry 5), there was low conversion and in addition to the product **2b**, the reaction tube had minor amount of physically different crystals than the crystals for **2b**. It single crystal X ray analysis showed unconjugated product **3b** (Figure 1). All attempts to isolate the compound **3b** in large quantity and its spectroscopic characterization were unsuccessful.

To check the scope of the reaction, a variety of thiols were tested. Apart from **1a** and **1b**, when 2-amino-4-chlorobenzenethiol **1c** was reacted with benzyl allenoate, thia-Michael adduct **2c** was formed in 93% yield. Again, no aza-Michael adduct, or *Z* isomer was detected from reaction mixture. Further aryl thiols including 2-bromobenzenethiol **1d** and 2,6-dichlorothiophenol **1e** and 2-naphthalenethiol **1f** under the same reaction conditions resulted in Michael adducts **2d**, **2e**



2b





3b

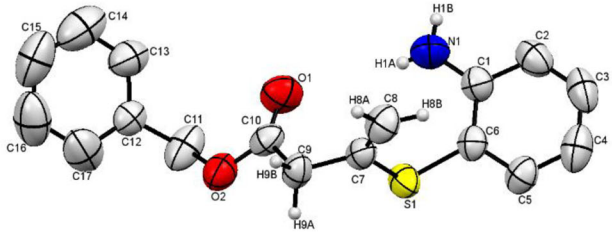
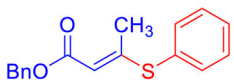


Figure 1. Single-crystal X-ray structures of **2 b** and **3 b**. The C-S bond and *E*-geometry of the product is confirmed. For details, see supporting information.

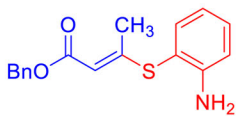


(150 mol%) **1a - 1l** **2a - 2l**

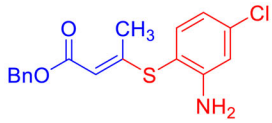
Yield[a], >20:1 *E/Z*



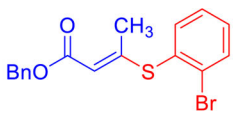
2a, 72%



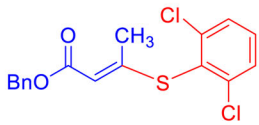
2b, 92%[b]



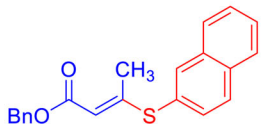
2c, 93%



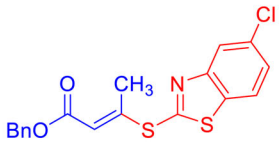
2d, 84%



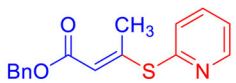
2e, 82%



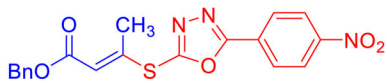
2f, 68%



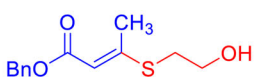
2g, 83%



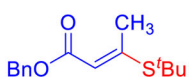
2h, 99%



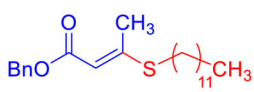
2i, 75%



2j, 92%



2k, 68%



2l, 82%

Scheme 2. Scope of reaction for thia-Michael addition. [a]Isolated yields, [b]Reaction with 150 mol% triethylamine.

and **2f** in 84%, 82% and 68% yields, respectively. Furthermore, heteroaryl thiols smoothly undergo thia-Michael addition to benzyl allenoate. Benzo[d]thiazole-2-thiol **1g** resulted in 83% of the thia-Michael adduct **2g**. 2-Pyridinethiol resulted in quantitative yield of the desire product **2h**. Similarly, highly functionalized 5-(4-nitrophenyl)-1,3,4-oxadiazole-2-thiol **1i** effectively reacts in thia-Michael fashion under the same reaction conditions to give **2i** in 75% yield.

Aliphatic thiols were also exposed to thia-Michael addition. 2-Mercaptoethanol afforded thia-Michael adduct **2j** in 92% yield. In case of **2j** thia-Michael adducts formation also demonstrated that free hydroxyl does not play a competitive role. Sterically demanding 2-methylpropane-2-thiol leads to product **2k** in 68% yield. A long chain aliphatic thiol dodecane-1-thiol efficiently resulted in adduct **2l** in 82% yield (Scheme 2). The reaction scope suggested that the reaction is general toward a variety of aliphatic, aryl, heteroaryl thiols in the presence of free hydroxyl and amine functionality.

3. Conclusion

In summary, we were able to obtain *E*-selective thia-Michael addition of aryl, heteroaryl and aliphatic thiols to benzyl allenoate. The reaction is facilitated by catalytic amount of triethylamine and is chemoselective toward thia-Michael addition in the presence of free amine and alcohol functional groups. The chemoselectivity and geometrical characterization were confirmed though single crystal X-ray analysis of **2b**. Additional characterization details for the X-ray structural studies are presented in the Supplemental Materials (Tables S1 and S2).

4. Experimental

Commercially available reagents were used without further purification unless necessary. Solvents were dried and distilled before their use. Reactions to be run in non-aqueous solvents were accomplished under an inert atmosphere of nitrogen. For Michael addition reaction of thiol, 13×100 mm sealed test tubes were used. To dry combined organic extracts, anhydrous sodium sulfate was used. Thin Layer Chromatography (TLC) was used for monitoring the reaction progress. UV visible spots were inspected under UV light at 254 nm. For visualization of UV inactive spots, different staining reagents such as potassium permanganate, phosphomolybdic acid (PMA), anisaldehyde or ninhydrin solution were used as appropriate. Flash column chromatography using silica gel was performed for purification of most of the products. Mass spectrometric (HRMS) experiments were carried out on Finnigan MAT-311A (Germany) mass spectrometer with (CI) ionization techniques. NMR spectra were obtained using a Bruker 300 NMR MHz spectrometer in deuterated solvents using TMS as internal reference, at 300 MHz (^{1}H NMR) and 75 MHz (^{13}C NMR). Chemical shifts are mentioned in delta (δ) units while coupling constants (J) values are in Hertz unit (Hz). The single crystal X-rays analysis was carried out with a Bruker Kapa APEX-II CCD diffractometer having a CCD detector set 40 mm from the crystal. From a sealed ceramic diffraction tube (SIEMENS) having graphite-monochromated Mo-Kα radiation, different intensities were determined. Then using Patterson and DIRDIF method, the structure of the compound was solved. The Supplemental Materials contains sample ^{1}H and ^{13}C NMR spectra for the products 2 (Figures S 1 – S 25).

4.1. Synthesis of substrate benzyl buta-2,3-dienoate

The benzyl buta-2,3-dienoate was synthesize in a slightly modified procedure to the previously reported method.[9] See the Supplemental Materials for the detailed experimental procedure.

4.2. General procedure for the addition of different thiols to allenoate

Oven-dried sealed tube was charged with benzyl buta-2,3-dienoate (123 μL, 0.75 mmol, 150 mol%), distilled triethylamine (19.2 μL, 0.1 mmol, 20 mol%) and nucleophile **1a-s** (0.5 mmol, 100 mol%). Freshly distilled THF (1 mL, 0.5 M) was added and purged with nitrogen. The mixture was placed in an oil bath at 70 °C and was allowed to stir for 24 h. Solvent was evaporated over dry silica gel and purification by flash column chromatography gave the desired product **2a-s**.

4.3. Benzyl (E)-3-(phenylthio)but-2-enoate (2a)

The reaction was performed with thiophenol **1a** (51 μL, 0.5 mmol, 100 mol%) and compound **2a** was obtained as brownish oil in 72% yield; ^{1}H NMR (300 MHz, $CDCl_3$): δ (ppm) 7.51 – 7.32 (m, 10H), 5.31 (*s*, 1H), 5.07 (*s*, 2H), 2.45 (s, 3H); ^{13}C NMR (75 MHz, $CDCl_3$): δ (ppm) 165.3, 160.7, 136.3, 135.5, 129.9, 129.7, 129.4, 128.5, 128.2, 128.0, 110.3, 65.5, 20.2; HRMS (CI) Calcd for $C_{17}H_{17}O_2S$ $[M+H]^+$: 285.0949; found, 285.0951; IR (neat) 3063, 3031, 2954, 2893, 1708, 1593, 1473, 1440, 1378, 1327, 1168, 1099, 1016, 747, 688, 581, 512 cm^{-1}

4.4. Benzyl (E)-3-(2-aminophenylthio)but-2-enoate (2b)

The reaction was performed with 2-aminothiophenol **1b** (62.5 mg, 0.5 mmol, 100 mol%) and compound **2b** was obtained as light brown crystals (mp 123 °C) in 92% yield; ^{1}H NMR (300 MHz, $CDCl_3$): δ (ppm) 7.37-7.20 (m, 7H), 6.77-6.71 (m, 2H), 5.29 (q, $J = 0.9$ Hz, 1H), 5.07 (s, 2H), 4.09 (broad s, 2H), 2.45 (d, $J = 0.9$ Hz, 3H); ^{13}C NMR (75 MHz, $CDCl_3$): δ (ppm) 165.0, 158.0, 148.7, 137.1, 136.1, 132.1, 128.4, 128.3, 128.0, 118.8, 115.5, 111.3, 109.5, 65.5, 19.9; HRMS (CI) Calcd for $C_{17}H_{18}NO_2S$ $[M+H]^+$: 300.1058; found, 300.1065; IR (neat) 3454, 3360, 3064, 3031, 2947, 2919, 2891, 1681, 1597, 1479, 1378, 1322, 1181, 1101, 1020, 927, 848, 743, 698 cm^{-1}

4.5. Benzyl (E)-3-(2-amino-4-chlorophenylthio)but-2-enoate (2c)

The reaction was performed with 2-amino-4-chlorothiophenol **1c** (100.8 mg, 0.5 mmol, 100 mol%) and compound **2c** was obtained in 93% yield; ^{1}H NMR (300 MHz, $CDCl_3$): δ (ppm) 7.33-7.23 (m, 6H), 6.76-6.69 (m, 2H), 5.30 (q, $J = 0.9$ Hz, 1H), 5.10 (s, 2H), 4.36 (broad s, 2H), 2.45 (d, $J = 0.9$ Hz, 3H); ^{13}C NMR (75 MHz, $CDCl_3$): δ (ppm) 164.9, 157.3, 149.5, 138.2, 137.9, 136.1, 128.4, 128.3, 128.1, 118.9, 115.1, 109.9, 109.8, 65.6, 19.7; HRMS (CI) Calcd for $C_{17}H_{17}ClNO_2S$ $[M+H]^+$: 334.0639; found, 334.0643; IR (neat) 3466, 3369, 2958, 2927, 2872, 1712, 1681, 1595, 1476, 1419, 1378, 1274, 1186, 1092, 1028, 945, 908, 839, 695, 597, 500 cm^{-1}

4.6. Benzyl (E)-3-(2-bromophenylthio)but-2-enoate (2d)

The reaction was performed with 2-bromothiophenol **1d** (94.5 mg, 0.5 mmol, 100 mol%) and compound **2d** was obtained in 84% yield; ^{1}H NMR (300 MHz, $CDCl_3$): δ (ppm) 7.73-7.61 (m, 1H), 7.62-7.59 (m, 1H), 7.36-7.28 (m, 7H), 5.26 (q, $J = 0.9$ Hz, 1H), 5.09 (s, 2H), 2.47 (d, $J = 0.9$ Hz, 3H); ^{13}C NMR (75 MHz, $CDCl_3$): δ (ppm) 164.9, 158.0, 137.5, 136.1, 134.0, 131.5, 130.8, 130.4, 128.46, 128.45, 128.2, 128.0, 110.8, 65.6, 19.9; HRMS (CI) Calcd for $C_{17}H_{16}BrO_2S$ $[M+H]^+$: 363.0054; found, 363.0057; IR (neat) 3062, 3031, 2950, 1707, 1600, 1446, 1376, 1332, 1257, 1168, 1096, 1019, 824, 747, 694, 595 cm^{-1}

4.7. Benzyl (E)-3-(2,6-dichlorophenylthio)but-2-enoate (2e)

The reaction was performed with 2,6-dichlorothiophenol **1e** (89.5 mg, 0.5 mmol, 100 mol%) and compound **2e** was obtained in 82% yield; ^{1}H NMR (300 MHz, $CDCl_3$): δ (ppm) 7.47-7.28 (m, 8H), 5.19 (q, $J = 1.2$ Hz, 1H), 5.09 (s, 2H), 2.48 (d, $J = 1.2$ Hz, 3H); ^{13}C NMR (75 MHz, $CDCl_3$): δ (ppm) 164.8, 156.2, 141.7, 136.1, 131.8, 129.0, 128.4, 128.3, 128.12, 128.10, 109.8, 65.7, 19.7; HRMS (CI) Calcd for $C_{17}H_{15}Cl_{12}O_2S$ $[M+H]^+$: 353.0170; found, 353.0176; IR (neat) 3558, 3065, 3033, 2953, 1709, 1602, 1555, 1424, 1377, 1333, 1169, 1099, 1022, 824, 777, 735, 695, 596 cm^{-1}

4.8. Benzyl (E)-3-(naphthalen-2-ylthio)but-2-enoate (2f)

The reaction was performed with 2-naphthalenethiol **1f** (80.1 mg, 0.5 mmol, 100 mol%) and compound **2f** was obtained in 68% yield; ^{1}H NMR (300 MHz, $CDCl_3$): δ (ppm) 8.04 (s, 1H), 7.89-7.82 (m, 3H), 7.57-7.49 (m, 3H), 7.33-7.27 (m, 5H), 5.37 (q, $J = 0.9$ Hz, 1H), 5.07 (s, 2H), 2.51 (d, $J = 0.9$ Hz, 3H); ^{13}C NMR (75 MHz, $CDCl_3$): δ (ppm) 165.0, 160.6, 136.2, 135.4, 133.7, 133.4, 131.4, 129.4, 128.5,4 128.2, 128.0, 127.9, 127.7, 127.4, 126.7, 126.6, 110.6, 65.4, 20; HRMS (CI) Calcd for $C_{21}H_{19}O_2S$ $[M+H]^+$: 335.1106; found, 335.1106; IR (neat) 3053, 3029, 2955, 2927, 2858, 1909, 1595, 1494, 1454, 1378, 1323, 1268, 1172, 1098, 1013, 866, 824, 750, 694, 583, 500 cm^{-1}

4.9. Benzyl (E)-3-(5-chlorobenzo[d]thiazol-2-ylthio)but-2-enoate (2g)

The reaction was performed with 5-chlorobenzo[d]thiazole-2-thiol **1g** (100.87 mg, 0.5 mmol, 100 mol%) and compound **2g** was obtained in 83% yield; ^{1}H NMR (300 MHz, $CDCl_3$): δ (ppm) 8.01 (d, $J = 1.8$ Hz, 1H), 7.72 (d, $J = 8.7$ Hz, 1H), 7.38-7.32 (m, 6H), 6.20 (q, $J = 1.2$ Hz, 1H), 5.15 (s, 2H), 2.56 (d, $J = 1.2$ Hz, 3H); ^{13}C NMR (75 MHz, $CDCl_3$): δ (ppm) 164.2, 161.4, 154.1, 152.9, 135.6, 135.1, 132.6, 128.4, 128.2, 128.1, 126.3, 122.9, 121.9, 118.4, 66.0, 20.4; HRMS (CI) Calcd for $C_{18}H_{15}ClNO_2S_2$ $[M+H]^+$: 376.0233; found, 376.0236; IR (neat) 3088, 3064, 3031, 2952, 2925, 1712, 1609, 1542, 1496, 1454, 1423, 1377, 1333, 1261, 1170, 1097, 987, 906, 865, 799, 732, 695, 578 cm^{-1}

4.10. Benzyl (E)-3-(pyridin-2-ylthio)but-2-enoate (2h)

The reaction was performed with 2-pyridinethiol **1h** (55.58 mg, 0.5 mmol, 100 mol%) and compound **2h** was obtained in quantitative yield; ^{1}H NMR (300 MHz, $CDCl_3$): δ (ppm) 8.62-8.60 (m, 1H), 7.69-7.63 (m, 1H), 7.47-7.44 (m, 1H), 7.34-7.32 (m, 5H), 7.26-7.20 (m, 1H), 5.85 (q, $J = 1.2$ Hz, 1H), 5.11 (s, 2H), 2.49 (q, $J = 1.2$ Hz, 3H); ^{13}C NMR (75 MHz, $CDCl_3$): δ (ppm) 164.7, 156.3, 153.6, 150.7, 137.3, 136.0, 128.3, 128.1, 128.0, 122.9, 114.9, 65.6, 20.5; HRMS (CI) Calcd for $C_{16}H_{16}NO_2S$ $[M+H]^+$: 286.0902; found, 286.0904; IR (neat) 3430, 3034, 2924, 2856, 1709, 1603, 1572, 1449, 1417, 1377, 1333, 1170, 1096, 1022, 738, 695 cm^{-1}

4.11. Benzyl (E)-3-(5-(4-nitrophenyl)-1,3,4-oxadiazol-2-ylthio)but-2-enoate (2i)

The reaction was performed with 5-(4-nitrophenyl)-1,3,4-oxadiazole-2-thiol **1i** (111.60 mg, 0.5 mmol, 100 mol%) and compound **2i** was obtained in 75% yield; ^{1}H NMR (300 MHz, $CDCl_3$): δ (ppm) 7.83 (d, $J = 8.4$ Hz, 2H), 7.35-7.26 (m, 5H), 6.72 (d, $J = 9$ Hz, 2H), 6.09 (q, $J = 0.9$ Hz, 1H), 5.14 (s, 2H), 2.48 (d, $J = 0.9$ Hz, 3H); ^{13}C NMR (75 MHz, $CDCl_3$): δ (ppm) 168.1, 164.1, 157.1, 151.0, 150.3, 135.7, 128.9, 128.8, 128.5, 128.3, 117.8, 114.6, 112.5, 66.2, 20.0; HRMS (CI) Calcd for $C_{19}H_{16}N_3O_5S$ $[M+H]^+$: 398.0811; found, 398.0815; IR (neat) 3447, 3406, 3349, 3224, 2927, 1713, 1601, 1480, 1444, 1334, 1291, 1174, 1071, 947, 837, 737, 696, 594 cm^{-1}

4.12. Benzyl (E)-3-(2-hydroxyethylthio)but-2-enoate (2j)

The reaction was performed with 2-mercaptoethanol **1j** (39 mg, 0.5 mmol, 100 mol%) and compound **2j** was obtained as light brown oil in 92% yield; ^{1}H NMR (300 MHz, $CDCl_3$): δ (ppm) 7.37-7.26 (m, 5H), 5.62 (s, 1H), 5.13 (s, 2H), 3.81 (t, $J = 6.3$ Hz, 2H), 2.97 (t, $J = 6.3$ Hz, 2H), 2.41 (s, 3H); ^{13}C NMR (75 MHz, $CDCl_3$): δ (ppm) 164.8, 158.6, 136.1, 128.5, 128.2, 128.1, 108.5, 65.6, 59.7, 34.1, 20.9; HRMS (CI) Calcd for $C_{13}H_{17}O_3S$ $[M+H]^+$: 253.0898;

found, 253.0896; IR (neat) 3367, 1707, 1591, 1373, 1336, 1172, 1105, 1025, 832, 740, 696 cm^{-1}

4.13. Benzyl (E)-3-(tert-butylthio)but-2-enoate (2k)

The reaction was performed with *tert*-butylthiol **1k** (45 mg, 0.5 mmol, 100 mol%) and compound **2k** was obtained as colorless oil in 68% yield; ^{1}H NMR (300 MHz, $CDCl_3$): δ (ppm) 7.38-7.34 (m, 5H), 5.92 (q, $J = 0.9$ Hz, 1H), 5.14 (s, 2H), 2.40 (d, $J = 0.9$ Hz, 3H), 1.47 (s, 9H); ^{13}C NMR (75 MHz, $CDCl_3$): δ (ppm) 165.0, 128.4, 128.3, 128.2, 128.1, 87.8, 79.4, 65.5, 46.9, 30.4, 22.2; HRMS (CI) Calcd for $C_{15}H_{21}O_2S$, 265.1262 $[M+H]^+$: found, 265.1265; IR (neat) 3067, 3033, 2959, 1969, 1712, 1589, 1497, 1455, 1327, 1242, 1150, 1081, 1024, 1000, 853, 735, 695, 592 cm^{-1}

4.14. Benzyl (E)-3-(dodecylthio)but-2-enoate (2l)

The reaction was performed with 1-dodecanthiol **1l** (101.2 mg, 0.5 mmol, 100 mol%) and compound **2l** was obtained in 82% yield; ^{1}H NMR (300 MHz, $CDCl_3$): δ (ppm) 7.40-7.25 (m, 5H), 5.55 (d, $J = 0.6$ Hz, 1H), 5.14 (s, 2H), 2.77-2.72 (s, 2H), 2.41 (d, $J = 0.6$ Hz, 3H), 1.69-1.26 (m, 20H), 0.88 (m, 3H); ^{13}C NMR (75 MHz, $CDCl_3$): δ (ppm) 164.9, 160.1, 136.4, 128.4, 128.1, 128.0, 107.7, 65.4, 31.8, 31.5, 29.6, 29.5, 29.4, 29.3, 29.0, 28.9, 28.8, 27.4, 22.6, 21.0, 14.1; HRMS (CI) Calcd for $C_{23}H_{37}O_2S$ $[M+H]^+$: 377.2516; found, 377.2514; IR (neat) 2921, 2852, 1710, 1593, 1455, 1377, 1336, 1258, 1170, 1104, 1025, 953, 832, 734, 695 cm^{-1}

5. Supplementary data

See the attached supporting information file with detailed experimental procedures and spectra for all compounds. The single crystal X-ray data associated with this article can be found at CCDC repository numbers, 1560208 and 1560209.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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